

EXPERIMENT 3

NAME: HARRISH L.S ROLL No. 240701178

CLI CODE:

```
import re
import math

COMMON_PASSWORDS = [
    "password", "123456", "12345678", "qwerty",
    "admin", "letmein", "welcome", "abc123"
]

def calculate_entropy(password):
    charset = 0

    if re.search(r"[a-z]", password):
        charset += 26

    if re.search(r"[A-Z]", password):
        charset += 26

    if re.search(r"[0-9]", password):
        charset += 10

    if re.search(r"^[a-zA-Z0-9]", password):
        charset += 32
```

```
if charset == 0:
```

```
    return 0
```

```
entropy = len(password) * math.log2(charset)
```

```
return entropy
```

```
def password_strength(password):
```

```
    feedback = []
```

```
    if password.lower() in COMMON_PASSWORDS:
```

```
        return "Very Weak  ", 0, ["Password is too common"]
```

```
    if len(password) < 8:
```

```
        feedback.append("Password length should be at least 8 characters")
```

```
    if not re.search(r"[A-Z]", password):
```

```
        feedback.append("Add at least one uppercase letter")
```

```
    if not re.search(r"[a-z]", password):
```

```
        feedback.append("Add at least one lowercase letter")
```

```
    if not re.search(r"[0-9]", password):
```

```
        feedback.append("Add at least one number")
```

```
if not re.search(r"^[a-zA-Z0-9]", password):  
    feedback.append("Add at least one special character")
```

```
entropy = calculate_entropy(password)
```

```
if entropy < 40:
```

```
    strength = "Weak 
```

```
    score = 25
```

```
elif entropy < 60:
```

```
    strength = "Medium 
```

```
    score = 50
```

```
elif entropy < 80:
```

```
    strength = "Strong 
```

```
    score = 75
```

```
else:
```

```
    strength = "Very Strong 
```

```
    score = 100
```

```
return strength, score, feedback
```

```
if __name__ == "__main__":
```

```
    pwd = input("Enter password: ")
```

```
    strength, score, tips = password_strength(pwd)
```

```
    print("\nPassword Strength:", strength)
```

```
print("Strength Score:", score, "%")
```

```
if tips:
```

```
    print("\nSuggestions to improve password:")
```

```
    for tip in tips:
```

```
        print("-", tip)
```

OUTPUT:

```
Enter password: Meow@1234  
  
Password Strength: Medium ⚠  
Strength Score: 50 %
```

GUI CODE:

```
import tkinter as tk
```

```
from tkinter import ttk, messagebox
```

```
import re
```

```
import math
```

```
COMMON_PASSWORDS = [
```

```
    "password", "123456", "12345678", "qwerty",
```

```
    "admin", "welcome", "abc123", "letmein"
```

```
]
```

```
def calculate_entropy(password):
```

```
    charset = 0
```

```
if re.search(r"[a-z]", password):
    charset += 26
if re.search(r"[A-Z]", password):
    charset += 26
if re.search(r"[0-9]", password):
    charset += 10
if re.search(r"^[a-zA-Z0-9]", password):
    charset += 32
if charset == 0:
    return 0
return len(password) * math.log2(charset)
```

```
def check_strength():
    password = password_entry.get()
    tips_box.delete(0, tk.END)

    if not password:
        messagebox.showwarning("Warning", "Enter a password")
        return

    if password.lower() in COMMON_PASSWORDS:
        strength_value.set("Very Weak )
        progress["value"] = 10
        tips_box.insert(tk.END, "Password is too common")
        return
```

```
tips = []
```

```
if len(password) < 8:
```

```
    tips.append("Use at least 8 characters")
```

```
if not re.search(r"[A-Z]", password):
```

```
    tips.append("Add an uppercase letter")
```

```
if not re.search(r"[a-z]", password):
```

```
    tips.append("Add a lowercase letter")
```

```
if not re.search(r"[0-9]", password):
```

```
    tips.append("Add a number")
```

```
if not re.search(r"^[a-zA-Z0-9]", password):
```

```
    tips.append("Add a special character")
```

```
entropy = calculate_entropy(password)
```

```
if entropy < 40:
```

```
    strength_value.set("Weak )
```

```
    progress["value"] = 25
```

```
elif entropy < 60:
```

```
    strength_value.set("Medium )
```

```
    progress["value"] = 50
```

```
elif entropy < 80:
```

```
    strength_value.set("Strong )
```

```
    progress["value"] = 75
```

else:

strength_value.set("Very Strong 🔒 ")

progress["value"] = 100

if tips:

for t in tips:

tips_box.insert(tk.END, t)

else:

tips_box.insert(tk.END, "Excellent password 👍 ")

root = tk.Tk()

root.title("Password Strength Checker")

root.geometry("520x420")

root.configure(bg="#f4f6f8")

root.resizable(False, False)

card = tk.Frame(root, bg="white", bd=0, relief="flat")

card.place(relx=0.5, rely=0.5, anchor="center", width=460, height=360)

tk.Label(card, text="🔒 Password Strength Checker",

font=("Segoe UI", 16, "bold"),

bg="white").pack(pady=15)

tk.Label(card, text="Enter your password",

font=("Segoe UI", 10),

```
fg="gray", bg="white").pack()
```

```
password_entry = tk.Entry(card, width=30, show="*",  
                           font=("Segoe UI", 12), relief="solid")
```

```
password_entry.pack(pady=10)
```

```
ttk.Button(card, text="Check Strength",  
           command=check_strength).pack(pady=10)
```

```
strength_value = tk.StringVar(value="Strength: —")
```

```
tk.Label(card, textvariable=strength_value,  
         font=("Segoe UI", 12, "bold"),  
         bg="white").pack(pady=8)
```

```
progress = ttk.Progressbar(card, length=300, maximum=100)  
progress.pack(pady=5)
```

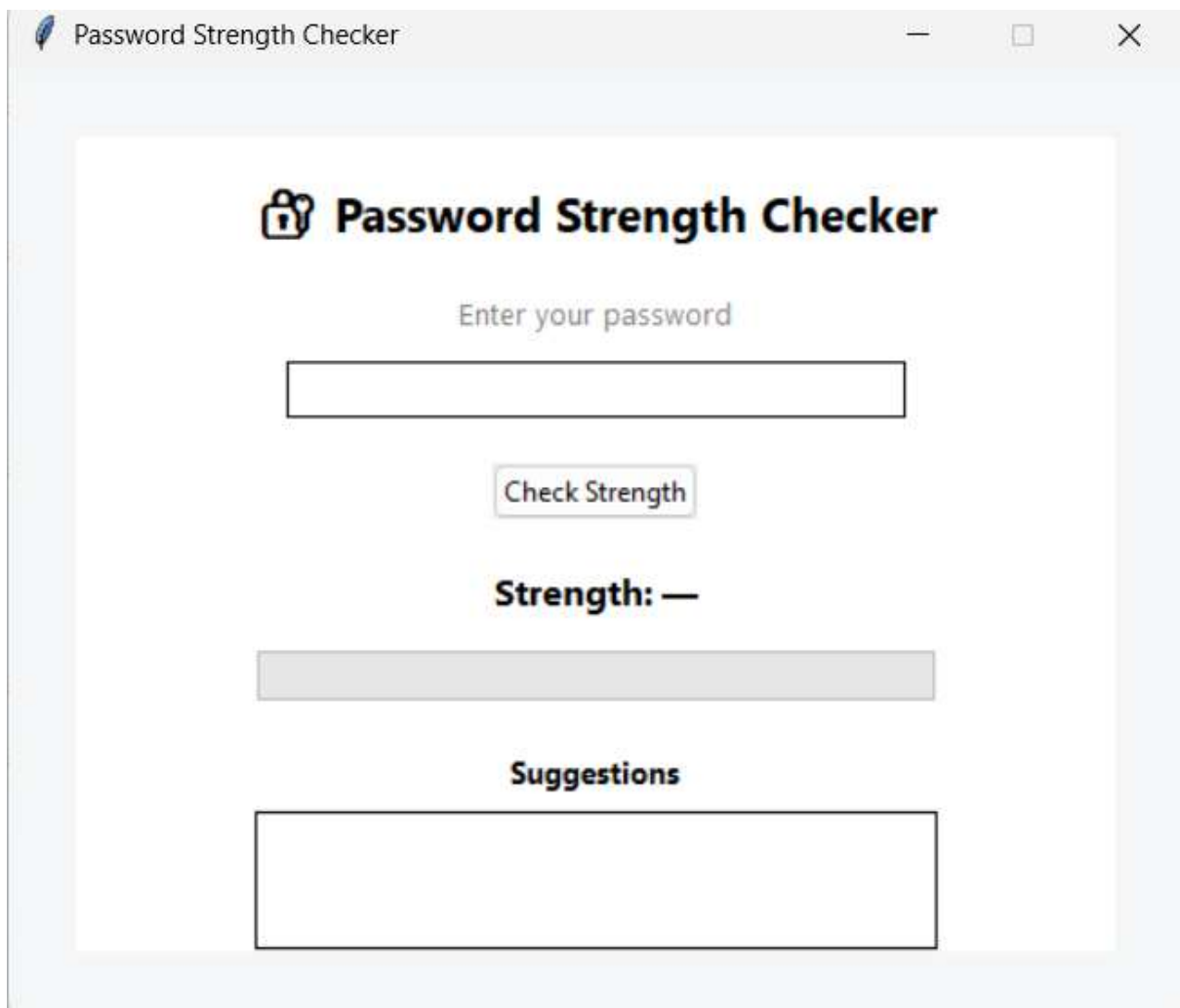
```
tk.Label(card, text="Suggestions",  
         font=("Segoe UI", 10, "bold"),  
         bg="white").pack(pady=(15, 5))
```

```
tips_box = tk.Listbox(card, width=50, height=6,  
                      font=("Segoe UI", 9),  
                      bd=1, relief="solid")
```

```
tips_box.pack()
```

```
root.mainloop()
```

OUTPUT:



The screenshot shows a window titled "Password Strength Checker". Inside the window, there is a header with a lock icon and the title "Password Strength Checker". Below the header, there is a text prompt "Enter your password" followed by a text input field. Under the input field is a button labeled "Check Strength". Below the button, the text "Strength: —" is displayed. Underneath this is a horizontal progress bar. At the bottom, the text "Suggestions" is shown above a larger text input field.

VUI CODE:

```
import re
import math

def calculate_entropy(password):
    charset = 0
    if re.search(r"[a-z]", password):
        charset += 26
    if re.search(r"[A-Z]", password):
        charset += 26
    if re.search(r"[0-9]", password):
        charset += 10
    if re.search(r"^[a-zA-Z0-9]", password):
        charset += 32
    if charset == 0:
        return 0
    return len(password) * math.log2(charset)

password = input("Speak your password (simulated): ")

entropy = calculate_entropy(password)

if entropy < 40:
```

```
    print("Password Strength: Weak")
elif entropy < 60:
    print("Password Strength: Medium")
elif entropy < 80:
    print("Password Strength: Strong")
else:
    print("Password Strength: Very Strong")
```

OUTPUT:

```
=====
Speak your password (simulated): meow@1234
Password Strength: Medium
|
```

USER SATISFACTION CODE:

```
def user_satisfaction():
    print("Rate your satisfaction for each interface (1 to 5)")

    try:
        cli = int(input("CLI Satisfaction (1-5): "))
        gui = int(input("GUI Satisfaction (1-5): "))
        vui = int(input("VUI Satisfaction (1-5): "))

        if not (1 <= cli <= 5 and 1 <= gui <= 5 and 1 <= vui <= 5):
            print("Error: Ratings must be between 1 and 5")
        return
```

```

print("\nUser Satisfaction Ratings")

print("CLI:", cli)

print("GUI:", gui)

print("VUI:", vui)


average = (cli + gui + vui) / 3

print("\nAverage Satisfaction Score:", round(average, 2))


except ValueError:

    print("Error: Please enter valid numbers only")


if __name__ == "__main__":

    user_satisfaction()

```

OUTPUT:

```

Rate your satisfaction for each interface (1 to 5)
CLI Satisfaction (1-5): 4
GUI Satisfaction (1-5): 4
VUI Satisfaction (1-5): 3


User Satisfaction Ratings
CLI: 4
GUI: 4
VUI: 3


Average Satisfaction Score: 3.67

```

